BCS-053	Multimedia Systems	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To learn and understand technical aspect of Multimedia Systems.
CO2	To understand the standards available for different audio, video and text applications
CO3	To Design and develop various Multimedia Systems applicable in real time.
CO4	To learn various multimedia authoring systems
CO5	To understand various networking aspects used for multimedia applications.

Detailed Syllabus

MODULE I

Introduction to Multimedia, Multimedia Information, Multimedia Objects, Multimedia in business and work. Convergence of Computer, Communication and Entertainment products Stages of Multimedia Projects, Multimedia hardware, Memory & storage devices, Communication devices, Multimedia software's, presentation tools, tools for object generations, video, sound, image capturing, authoring tools, card and page based authoring tools.
Multimedia Building Blocks Text, Sound MIDI, Digital Audio, audio file formats, MIDI underwindows environment Audio & Video Capture.

MODULE II

Data Compression Huffman Coding, Shannon Fano Algorithm, Huffman Algorithms, Adaptive Coding, Arithmetic Coding Higher, Order Modeling. Finite Context Modeling, Dictionary basedCompression, Sliding Window Compression, LZ77, LZW compression, Compression, Compression ratio loss less & lossy compression.

MODULE III

Speech Compression & Synthesis Digital Audio concepts, Sampling Variables, Loss less compression of sound, loss compression & silence compression. Images Multiple monitors, bitmaps, Vector drawing, lossy graphic compression, image file formatic animations Images standards, JPEG Compression, Zig Zag Coding, Multimedia Database. Video: Video representation, Colors, Video Compression, MPEG standards, MHEG Standard Video Streaming on net, Video Conferencing, Multimedia Broadcast Services.

Text Books:

1. Tay Vaughan "Multimedia, Making IT Work" Osborne McGraw Hill.
2. Buford "Multimedia Systems" Addison Wesley.
3. Agrawal & Tiwari "Multimedia Systems" Excel.
4. Mark Nelson "Data Compression Book" BPB.
5. David Hillman "Multimedia technology and Applications" Galgotia Publications.
6. Rosch "Multimedia Bible" Sams Publishing.
7. Sleinreitz "Multimedia System" Addison Wesley.
8. James E Skuman "Multimedia in Action" Vikas.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Developed understanding of technical aspect of Multimedia Systems.
CO2	Understand various file formats for audio, video and text media.
CO3	Develop various Multimedia Systems applicable in real time.
CO4	Design interactive multimedia software.
CO5	Apply various networking protocols for multimedia applications.